COMPANY OPERATING SYSTEM · VERSION 1.0 · 3 NOV 2026

Mobile Application Development Company Operating System

One repeatable system to research, design, build, test, publish, price, sell and maintain profitable mobile apps — with SOPs, templates, approval gates, costing formulas and a fully worked example.

Document control	
Owner	CTO / Managing Director
Approver	Founders (Gate owner list in §2)
Status	Draft v1.0 for founder review
Currency / units	INR (₹), Indian digit grouping; hours; ex-GST unless stated
Companion file	build/model.py recomputes every number in this document

Read this first - what is verified and what is not
Prices of cloud services, SaaS tools, SMS/email and store fees were **not** verified against live official pages while preparing this edition (the research environment could not reach them). Every such figure is labelled **Planning** or **Last known** and must be replaced from the official page or calculator, with the access date, at **Gate 8**. No market statistics are quoted; Phase 1 requires every statistic to carry a source URL, access date and confidence level.

★ Executive summary

₹15.01 L
Expected build labour (sample app)

1,747 h
Expected effort (PERT)

₹31,10,289
Target custom-build price (client #1)

70 customers
SaaS operating break-even

93%
Gross margin (cost to serve)

₹78,516/mo
Operating profit at 100 customers

This document is the standing instruction set for the whole company. It does five things:

1. **Intake** (§1): captures company facts and sets explicit defaults where facts are still missing.
2. **Lifecycle and 11 approval gates** (§2): nobody starts a phase without the gate signed.
3. **Phase SOPs 1-7, 10-13** (§3-§9, §12-§15): owner, inputs, procedure, output, exit criteria, templates.
4. **Costing and pricing engine** (§10-§11): formulas plus a complete worked example for one sample app, *SiteDiary*.
5. **People and portal** (§16-§17): role responsibilities, weekly report, and the spec for the internal project portal.

1 Company intake (First task)

Gate 0

> **Task 0 - Intake**
> Fill the third column of the table at the kickoff meeting. Values shown are **working assumptions** so the system can be demonstrated end to end; replace them and re-run `model.py` / the calculator in §10.

Question	Working assumption	Note / where used
Company name	[To be provided]	Placeholder: *Your Company Pvt Ltd*
Business model	Hybrid: own SaaS products + paid client builds	Own products give recurring revenue; client work funds the runway
Employees	6 (PM, UX, Flutter, Backend, QA, DevOps/DB) + part-time support	Used for rates in §10
Technology budget (cash, non-labour)	₹3,00,000 for first product	Sample app non-labour cash need: ₹2,08,160 (§10-D)
Target industries	Construction & field services (first); then retail, clinics	Choose one wedge industry per product
Target customers	Small contractors / site supervisors in India (5-50 staff)	Validate in Phase 1 interviews
Preferred technology	Flutter + NestJS/TypeScript + PostgreSQL + Firebase Cloud Messaging	Decided per project by the §5 matrix, not by default
Own vs client apps	Both	Pricing models for both are in §11
Initial project	SiteDiary - offline-first daily site report app	Fictional sample used throughout

Intake questionnaire (copy for every new project)

- ☐ Who pays: our own product, a client, or an enterprise buyer?
- ☐ Which of Android, iOS, web is genuinely required? (Default: ask if a PWA suffices.)
- ☐ Is offline use required? Which operations? Conflict rule?
- ☐ Which personal / sensitive data is collected? (Triggers DPDP Act 2023 review.)
- ☐ Payment, SMS, maps or AI features needed?
- ☐ Budget ceiling and deadline? Who signs each gate?
- ☐ Who owns source code and customer data after delivery?

2 Lifecycle and 11 approval gates

Market research → G1 → Requirements + I/O freeze → G2 → Tech + architecture → G3 → UI/UX → G4 → Repo + env setup → G5 → Build (mobile + backend) → G6 → QA + security → G7 → Costing + pricing → G8 → Production deploy → G9 → Store publishing → G10 → Showcase + sales → Handover → G11 → Maintenance

Gate	Name	Evidence prepared by	Approver	Minimum exit criteria
G1	Market validation	Business Researcher / PM	Founder	Research report with sourced evidence; ≥10 customer interviews or equivalent; go / pivot / stop decision
G2	Requirements & scope freeze	PM	Founder + customer (if client)	PRD, screen I/O spec, API-DB map signed; change-control active
G3	Architecture & technology	Tech lead	CTO	Decision matrix, 11 architecture docs, threat model, ADRs
G4	UI/UX approval	UX Designer	PM + customer	Clickable prototype, usability test ≥5 users, accessibility check
G5	Repo & dev setup	DevOps	Tech lead	Protected main, CI green, 3 environments, secrets policy, .env.example
G6	MVP development review	Tech lead	CTO + PM	All frozen screens built; demo against spec; ≥80% unit coverage on business logic
G7	QA & security	QA owner	CTO	Zero open Sev-1/2, OWASP MASVS checklist done, device matrix passed
G8	Costing & pricing	PM + Finance	Founders	Verified provider prices (dated), cost model, price floor, break-even
G9	Production deployment	DevOps	CTO	Restore drill passed, monitoring + alerts live, rollback tested
G10	Store submission & release	Release manager	CTO + PM	Store checklists complete, privacy declarations match SDK audit
G11	Customer handover	PM	Founder + customer	Manuals, training, credentials transfer, support contract signed

> **Rule**
> No employee may start work belonging to a later phase until the preceding gate is recorded as **Approved** in the portal (§17) with approver name, date and evidence links. Rejected or revision-requested gates return the work to the owner with written reasons.

3 Phase 1 - Market research & validation

Gate 1

OWNER
Business Researcher (PM backup)

INPUTS
Idea one-pager; industry & customer hypotheses

OUTPUT
Market Research Report (PDF)

EXIT GATE
Gate 1

PROCEDURE

1. Write the problem statement in one sentence and name the buyer, user and payer.
2. Collect competitor data from Google Play, Apple App Store, Product Hunt, G2/Capterra, Reddit, Google Trends and official sites. Log each fact in the evidence register below.
3. Read the 1- and 2-star reviews of the top five competitors; tag complaints (missing feature, price, bug, support).
4. Run ≥10 customer interviews (script: problem, current workaround, willingness to pay, budget owner).
5. Size the market **bottom-up**: TAM = number of potential customers × annual revenue per customer; SAM = reachable segment; SOM = realistic 3-year capture. Cite the source of every count.
6. Choose the monetisation model: paid, subscription, freemium or enterprise; justify with competitor pricing and interview evidence.
7. Rate customer-acquisition difficulty (channels, cost, sales cycle) and state a go / pivot / stop recommendation.

Evidence register (mandatory format)

ID	Finding	Source URL	Access date	Confidence (H/M/L)	Used for
E-001	*(example row)* Competitor X lists price ₹___/month	https://...	YYYY-MM-DD	H - official page	Pricing benchmark
E-002	*(example row)* 38 of 120 reviews mention no offline mode	https://... (store page)	YYYY-MM-DD	M - own tagging	Feature gap

Bottom-up sizing formula

```
TAM = N(potential customers) × ARPA(₹/yr)
SAM = TAM × share reachable via our channels and language/geography
SOM = SAM × realistic 3-yr share (state the basis; ≤5% unless evidence says otherwise)
```

> **No invented statistics**
> If a number cannot be sourced, write **Unknown - research needed**. Reports with unsourced figures are rejected at Gate 1. Use government sources (e.g. MeitY, ministry reports) and official competitor pages as primary evidence.

4 Phase 2 - Requirements & I/O freeze

Gate 2

OWNER
Product Manager

INPUTS
Approved market report

OUTPUT
PRD + Screen Specification + I/O Freeze document

EXIT GATE
Gate 2

PROCEDURE

1. Write the PRD: objective, users, roles and permissions, modules, navigation, journeys, offline/online scope, integrations, SDKs, security, performance, device support, accessibility, maintenance.
2. For **every screen** complete the Screen I/O sheet (below). One sheet per screen, stored in `docs/requirements/screens/`.
3. Build the screen → API → database map and the failure-behaviour table.
4. Review with UX, tech lead and QA (testability of each rule). Resolve all open questions.
5. Freeze: tag the PRD `prd-v1.0-frozen`. Any later change requires a Change Request (CR).

Screen I/O sheet (example filled)

Field	Content
Screen ID / name	S-014 Daily Report - Attendance
Purpose / role	Record who worked today / Supervisor
Inputs & types	worker_id (picker), status (enum: present, absent, half), hours (decimal 0-16), note (text ≤200)
Validation	status required; hours required if present; hours step 0.5; no duplicate worker per date
Buttons → actions	Save → local write + sync queue; Submit report → locks day, enqueues upload
API → DB	POST /v1/reports/{id}/attendance → INSERT attendance(...) ; GET same → SELECT
Output / errors	Toast 'Saved offline'; E-4101 'Hours must be 0-16'
Loading / offline	Skeleton rows; fully usable offline, badge 'Pending sync'

Mandatory failure behaviours

Situation	Required behaviour (default - override in PRD)
Slow internet	Show cached data immediately; spinner after 400 ms; timeout 15 s then 'Retry' without losing input
No internet	Offline banner; allowed offline ops list (§ PRD); writes queue locally
API failure (5xx)	Exponential backoff (2,4,8,16 s, max 5); user sees non-technical message + request id
Session expiry (401)	Silent refresh once; else save draft locally, route to login, restore draft after login
Invalid login	Generic message, no hint which field; lockout 5 attempts / 15 min
Permission denied	Explain why the permission is needed; offer settings link; degrade gracefully
App restart / rotation	Restore last route and unsaved form from local draft; layouts handle rotation where supported
Interrupted upload	Resumable chunked upload; resume on reconnect; checksum verify
Failed notification	Never the sole channel; in-app inbox mirrors every push; token refresh on rotation

Change-request form

```
CHANGE REQUEST  CR-<project>-<nnn>
Requested by / date:          Reason & business value:
Affected screens / APIs / tables:
Effort by role (hours) and cost (use §10 rates):
Schedule impact (days):       Risk:
Decision: Approved / Rejected / Deferred      Approver + date:
```

5 Phase 3 - Technology & architecture

Gate 3

OWNER
Tech Lead

INPUTS
Frozen PRD

OUTPUT
ADR + architecture pack

EXIT GATE
Gate 3

PROCEDURE

1. Shortlist options; score with the matrix below; record the decision as an ADR.
2. Challenge the premise: can a responsive web app / PWA meet the requirement? If yes, recommend it.
3. Produce the 11 architecture documents; threat-model every trust boundary.
4. Review cost implications with PM before approval.

Technology decision matrix

Criterion	Weight	Example scoring guide (1-5)
Dev cost	20%	Flutter 4 / Native 2 / RN 4 / PWA 5
Performance	10%	Native 5 / Flutter 4 / RN 3 / PWA 3
Platform-specific features	10%	Native 5 / Flutter 3 / RN 3 / PWA 2
Maintenance cost	15%	Flutter 4 / Native 2 / RN 4 / PWA 5
Offline capability	10%	Native 5 / Flutter 4 / RN 4 / PWA 3
Team experience	15%	Score from actual skills
Security	5%	Native 5 / others 4 (equal controls required)
Scalability / vendor risk	15%	Score per backend/cloud choice
Weighted score	**100%**	**Σ weight × score; ties → lowest maintenance cost**

Default stack and exceptions

Layer	Default choice	Choose something else when
Mobile	Flutter (one codebase, Android+iOS)	Heavy native APIs/AR/BLE → Native; team is React-only → React Native; no device features → PWA
Backend	NestJS (TypeScript) REST + OpenAPI	.NET/Python if the team's depth is there; Supabase/Firebase for tiny MVPs
Database	PostgreSQL; SQLite (Drift) on device; Redis only if caching proven needed	Firestore if data is document-shaped and offline-sync by vendor is acceptable
Cloud	Managed container + managed PG in an India region	AWS/Azure/GCP by customer mandate or credits; check data-residency needs
Notifications	FCM (Android + iOS via APNs bridge)	Direct APNs only if FCM limits hit

Required architecture documents

- ☐ Application architecture (C4 context + container)
- ☐ Frontend architecture (layers, state, navigation)
- ☐ Backend architecture (modules, queues)
- ☐ Database architecture (ER diagram, indexes, migrations)
- ☐ API architecture (REST, versioning, error model)
- ☐ Authentication (OTP/password, JWT + refresh, device binding)
- ☐ Notification architecture
- ☐ Offline sync architecture (outbox, conflict policy)
- ☐ File storage (signed URLs, lifecycle)
- ☐ Threat model (STRIDE; MASVS mapping)
- ☐ Cloud infrastructure diagram

> **Offline is a design decision**
> Define per entity: allowed offline operations, local retention, sync order, and conflict rule (e.g. *server-wins for master data; append-only for reports; last-writer-wins only with timestamp + device id, and user-visible conflict screen for edits*). Do not sync everything by default.

6 Phase 4 - UI/UX & prototype

Gate 4

OWNER
UX Designer

INPUTS
Frozen PRD and architecture

OUTPUT
Figma prototype + design tokens + component library

EXIT GATE
Gate 4

PROCEDURE

1. Journey maps → information architecture → screen inventory (same IDs as PRD).
2. Low-fi wireframes, review with PM; then high-fi in Figma using a token-based design system.
3. Include loading, empty, error, permission-prompt and offline states for every screen.
4. Build clickable prototype; test with ≥5 representative users; log issues and fix Sev-1/2.
5. Accessibility pass: contrast ≥4.5:1, touch targets ≥48dp (Android) / 44pt (iOS), dynamic text, screen-reader labels.
6. Export tokens and component specs for developers; get PM + customer sign-off.

Standard	Rule
Platform conventions	Material guidance on Android, Human Interface Guidelines on iOS (Flutter adaptive widgets where it matters)
Tokens	Colour, type scale, 4/8 pt spacing, radius, elevation defined once; no hard-coded values in screens
Originality	Never copy competitor UI; original brand name, colours, icon and tone
Dark mode	Only if in PRD; tokens must support it from day one

7 Phase 5 - Environment & repository

Gate 5

OWNER
DevOps Engineer

INPUTS
Approved architecture

OUTPUT
Configured repo + env

EXIT GATE
Gate 5

PROCEDURE

1. Install and verify tools on every workstation (table).
2. Create repo from the company template; apply branch protection.
3. Create three environments with separate credentials; wire CI/CD.
4. Dry-run: fresh clone → build → test → deploy to dev in under 30 minutes (onboarding test).

Tool installation checks

Tool	Check command	Expected
Git	git --version	≥ 2.40
Flutter / Dart	flutter --version && flutter doctor -v	No × for chosen targets
Node.js (LTS)	node -v && npm -v	Active LTS
Android Studio / SDK	flutter doctor --android-licenses	Licences accepted
Xcode (macOS only)	xcodebuild -version	Needed for iOS build/sign
PostgreSQL client	psql --version	Matches server major
Postman / Bruno	n/a	Collection in docs/api

Repository structure

```
mobile-app/
  mobile/        lib/{core,config,services,models,repositories,features,widgets,navigation,state}
                 assets/  test/  integration_test/
  backend/       src/  tests/  migrations/
  admin-dashboard/        (only if PRD requires)
  docs/          requirements/ architecture/ api/ database/ ui-ux/ deployment/ testing/
  infra/         .github/workflows/   .env.example   .gitignore   README.md   CONTRIBUTING.md
```

GitHub standards

- ☐ Private repository per product; org-level 2FA enforced
- ☐ main protected: PR required, ≥1 review, status checks required, no force-push
- ☐ Branches: feature/-, fix/..., release/x.y.z; squash-merge
- ☐ Commits: Conventional Commits (feat:, fix:, docs:, chore:) + task id
- ☐ Actions: lint, unit tests, build APK/AAB/IPA debug, dependency & secret scan, CodeQL
- ☐ Dependabot enabled; release tags vX.Y.Z; CHANGELOG generated
- ☐ Environments dev / staging / production with separate secrets, databases and API URLs
- ☐ Secrets only in GitHub/Cloud secret stores; `.env.example` lists names only
- ☐ Pre-commit secret scan; signing keys stored offline + in vault, never in repo

8 Phase 6 - Development

Gate 6

OWNER
Tech Lead + Developers

INPUTS
Frozen PRD, approved design, ready repo

OUTPUT
Working MVP

EXIT GATE
Gate 6

PROCEDURE

1. Slice work by module; each task = one PR linked to PRD screen ID.
2. Test-first for business rules; mandatory code review; demo to PM every sprint.
3. Never build screens not in the PRD; raise a CR instead.
4. Track estimate vs actual hours per task in the portal.
5. At Gate 6, demo every frozen screen against its I/O sheet.

A. Mobile frontend scope

- Splash, onboarding, registration, login, forgot password
- Dashboard, navigation, core modules per PRD
- Forms, search/filter, reports
- Notifications, profile, settings, help, logout

B. Backend scope

- AuthN/AuthZ (RBAC), user management, business logic, REST endpoints + OpenAPI
- Database access via migrations only; audit trail on writes
- File management (signed URLs), notification service, payments if required
- Structured logging with request ids; central error model

C. Offline standard

Aspect	Standard
Local DB	SQLite via Drift; schema versioned and migrated
Outbox	Every offline write = row in outbox (id, entity, op, payload, created_at, attempts)
Retry	Exponential backoff + jitter, max attempts, then 'needs attention' list
Conflict	Rule per entity (see §5 callout); conflicts logged and surfaced
Indicators	Offline banner, per-item 'pending' badge, 'last synced' timestamp
Idempotency	Client-generated UUID + Idempotency-Key header so retries never duplicate

D. Push notifications

- Ask permission in context (not at first launch); explain value
- Store device tokens per user-device; delete on logout; refresh on rotation
- Handle invalid-token errors by pruning; never rely on push as the only channel
- Per-category preferences and quiet hours; no personal data in payloads

E. Admin dashboard (if PRD requires)

- User management, roles, settings
- Analytics and reports

- Subscription management, notifications, support tools
- Audit-log viewer; every action by role-based access

9 Phase 7 - QA & security

Gate 7

OWNER
QA Engineer (owner)

INPUTS
Build candidate + PRD

OUTPUT
QA & Security Report

EXIT GATE
Gate 7

PROCEDURE

1. Write test cases from each I/O sheet (traceability matrix: requirement → test → result).
2. Run functional, device, network, performance, security tests.
3. Automate: unit, widget, integration, API and a smoke E2E suite in CI.
4. Report defects by severity; Sev-1/2 must be zero to pass.
5. Publish the QA & Security Report.

Device and network matrix

Dimension	Minimum matrix (replace with analytics-based list at G7)
Android	API levels covering the top versions by your analytics, incl. one low-RAM (≤3 GB) device
iOS	Current and previous major iOS; smallest and largest iPhone
Network	Wi-Fi, 4G, 2G-throttled, airplane mode, network switching, request interruption
Tablets	Only if in PRD

Performance targets

Metric	Target (default)
Cold start	≤ 3 s on mid-range device
Screen load (cached)	≤ 1 s
API p95	≤ 500 ms at planned load
Crash-free sessions	≥ 99.5%
ANR rate (Android)	< 0.47% (Play bad-behaviour threshold; verify current value)
APK/AAB size	Track; justify >50 MB

Security checklist (OWASP MASVS / MASTG)

- ☐ Auth & session handling (token expiry, revocation, brute-force)
- ☐ Tokens in Keystore/Keychain, not plain storage
- ☐ API authorisation per object (IDOR tests)
- ☐ TLS only, certificate validation, consider pinning with rotation plan
- ☐ No secrets / PII in logs, screenshots or crash reports
- ☐ Deep links & WebViews validated
- ☐ Third-party SDK review (data collected vs privacy declaration)
- ☐ Data deletion works end to end
- ☐ Permissions minimal
- ☐ Map results to OWASP MASVS levels; test with MASTG procedures

Severity	Definition	Gate rule
Sev-1	Crash, data loss, security hole, blocked core flow	Zero allowed
Sev-2	Major feature wrong, no workaround	Zero allowed
Sev-3	Minor with workaround	Documented, scheduled
Sev-4	Cosmetic	Backlog

10 Phase 8 - Cost calculation (worked example: SiteDiary)

Gate 8

Sample product: **SiteDiary** - an offline-first Flutter app (Android + iOS) where site supervisors record attendance, material receipts, issues and photos, and the owner receives a PDF daily report; NestJS + PostgreSQL backend, web admin dashboard, subscription billing. Scope = tasks T01-T23 below.

8.0 Assumptions register (change here first)

ID	Assumption	Value	Status	Note
A-01	Employee overhead loading	25%	Planning	Equipment, seats, workspace, management, statutory
A-02/03	Paid hours / utilisation	160 h × 75% = 120 productive h/mo	Planning	Leave, meetings, idle time excluded
A-04	Risk classes (O / P multipliers)	Low .85/1.25 · Med .80/1.40 · High .75/1.80	Planning	Calibrate from actual-vs-estimate history quarterly
A-08	Reuse factor on reusable tasks (client #2)	40% of original effort	Planning	Measure after first reuse
A-10	USD→INR	₹90	Planning	Replace with RBI reference rate on the day
A-11	Backend instance cost / capacity	₹3,500/mo · 150 req/s	Planning	Capacity must be proven by load test
A-12	Storage / egress / SMS / email	₹2.0/GB-mo · ₹8.0/GB · ₹0.25/SMS · ₹0.08/email	Planning	Replace from provider calculators at Gate 8
A-13	Photo size / retention	0.6 MB after compression · 12 months hot	Planning	Measure in pilot
A-14	Payment gateway fee	2.0% of billed amount	Planning	Verify gateway pricing + GST on fee
A-15	Corporate income tax	25.17% (new regime company rate incl. surcharge+cess)	Verify with CA	Shown below operating profit only
A-16	GST on subscriptions	18% collected, not revenue	Verify with CA	Pass-through; shown separately
A-17	Churn / CAC	2.5% per month · ₹9,000 per customer	Hypothesis	Validate in pilot; see sensitivity
A-18	Plan mix / annual share	50/35/15% Starter/Pro/Business · 30% annual (10 months paid)	Hypothesis	Validate with pilot customers

8.1 Fully loaded hourly rates

```
Fully loaded hourly cost = Monthly CTC × (1 + overhead%) ÷ (paid hours × utilisation)
Example (Flutter): 85,000 × 1.25 ÷ 120 = 885.42 → rounded ₹885/h
```

Role	Monthly CTC	Loaded / month	Divisor	Rate / hour
Product Manager / Business Researcher	₹90,000	₹1,12,500	÷ 120 h	₹938
UI/UX Designer	₹70,000	₹87,500	÷ 120 h	₹729
Flutter Developer (Android+iOS)	₹85,000	₹1,06,250	÷ 120 h	₹885
Backend Developer	₹90,000	₹1,12,500	÷ 120 h	₹938
QA Engineer	₹50,000	₹62,500	÷ 120 h	₹521
DevOps / Database Engineer	₹90,000	₹1,12,500	÷ 120 h	₹938
Security Specialist (external)	Outsourced	-	₹2,500/h fixed	₹2,500
Customer support executive (part-time)	₹35,000	₹43,750	÷ 120 h	₹365

> **Salaries are illustrative**
> CTC values are placeholders chosen to demonstrate the method. Use real payroll data; never publish individual salaries outside Finance.

8.2 Hours and labour cost - optimistic / expected / pessimistic

```
PERT expected hours E = (O + 4·M + P) ÷ 6 ; O = M × optimistic factor ; P = M × pessimistic factor (by risk class)
Task cost = Σ over roles (hours × role rate) ; scenario cost = most-likely cost × (scenario hours ÷ M)
```

ID	Task	C/R	Risk	Likely h	Opt h	Exp h	Pess h	Optimistic ₹	Expected ₹	Pessimistic ₹
T01	Market research & validation	R	Low	40	34	41	50	₹31,892	₹38,145	₹46,900
T02	Requirements, PRD & I/O freeze	C	Med	80	64	83	112	₹57,936	₹74,834	₹1,01,388
T03	UI/UX design, design system & prototype	C	Med	130	104	134	182	₹78,821	₹1,01,810	₹1,37,936
T04	Architecture & technology selection	R	Low	44	37	45	55	₹34,541	₹41,313	₹50,795

ID	Task	C/R	Risk	Likely h	Opt h	Exp h	Pess h	Optimistic ₹	Expected ₹	Pessimistic ₹
T05	Repository, CI/CD & environments	R	Low	36	31	37	45	₹28,432	₹34,008	₹41,812
T06	Database design & migrations	C	Low	36	31	37	45	₹28,703	₹34,331	₹42,210
T07	Authentication, roles & user management	R	Med	72	58	74	101	₹50,173	₹64,807	₹87,802
T08	Project/site & worker master modules	C	Low	78	66	79	98	₹56,843	₹67,989	₹83,592
T09	Daily report: attendance, materials, issues	C	High	140	105	153	252	₹89,055	₹1,29,624	₹2,13,732
T10	Offline storage & sync engine	R	High	144	108	157	259	₹89,685	₹1,30,542	₹2,15,244
T11	Resumable photo upload & file storage	R	Med	62	50	64	87	₹42,669	₹55,114	₹74,670
T12	Report PDF generation & sharing	C	Med	48	38	50	67	₹32,672	₹42,201	₹57,176
T13	Push notifications (FCM/APNs)	R	Low	32	27	33	40	₹22,756	₹27,218	₹33,465
T14	Admin dashboard (web)	R	Med	120	96	124	168	₹81,494	₹1,05,264	₹1,42,615
T15	Subscription billing & payments	R	High	58	44	63	104	₹37,198	₹54,144	₹89,276
T16	Third-party integrations (OTP, email, maps)	R	Low	22	19	22	28	₹17,180	₹20,549	₹25,265
T17	Functional, device & network testing	C	Med	168	134	174	235	₹85,018	₹1,09,814	₹1,48,781
T18	Security review & remediation	R	Med	64	51	66	90	₹77,338	₹99,894	₹1,35,341
T19	Performance tuning & bug-fix reserve	C	Med	96	77	99	134	₹62,336	₹80,517	₹1,09,088
T20	Production deployment & store submission	R	Med	72	58	74	101	₹53,350	₹68,911	₹93,363
T21	Documentation & user/admin manuals	C	Low	48	41	49	60	₹32,599	₹38,991	₹47,940
T22	Customer training & pilot onboarding	C	Low	28	24	28	35	₹19,489	₹23,310	₹28,660
T23	Project management & coordination	C	Low	60	51	61	75	₹47,838	₹57,218	₹70,350
	TOTAL			**1,678**	**1,346**	**1,747**	**2,422**	**₹11,58,018**	**₹15,00,549**	**₹20,77,404**

C = customer-specific work, R = reusable company IP (auth, sync engine, uploads, billing, admin, CI). Hours are assigned once to the role that performs them - one person covering two roles must not be double counted (a developer who also tests appears once, under the role with the larger share).

8.3 Hours by role (most-likely)

Role	Most-likely hours	Rate/h	Cost at most-likely
Product Manager / Business Researcher	218	₹938	₹2,04,484
UI/UX Designer	132	₹729	₹96,228
Flutter Developer (Android+iOS)	490	₹885	₹4,33,650
Backend Developer	430	₹938	₹4,03,340
QA Engineer	278	₹521	₹1,44,838
DevOps / Database Engineer	106	₹938	₹99,428
Security Specialist (external)	24	₹2,500	₹60,000
Total (most-likely)	**1,678**		**₹14,41,968**

Live labour & custom-price calculator (web version only; same formulas as the tables)

Rates: PM ₹938/h · UX ₹729/h · FL ₹885/h · BE ₹938/h · QA ₹521/h · DO ₹938/h · SE ₹2,500/h

₹15,00,549	₹20,77,404	₹21,77,202	₹31,10,289
Expected labour	Pessimistic labour	Total cost	Target price (client #1)

8.4 Software and licence cost

Item	Class	₹ / month	Basis
VS Code, Android Studio, Flutter SDK, Xcode, Git	Free	Free	Official downloads
GitHub (private repos, Actions) - 6 seats	Paid	₹2,400	Planning - verify github.com/pricing
Figma (2 design seats)	Paid	₹2,600	Planning - verify figma.com/pricing

Item	Class	₹ / month	Basis
Firebase Crashlytics / Analytics / FCM	Free	Free	Verify Firebase pricing page
Sentry or similar (optional)	Optional	₹2,400	Planning
Postman / Bruno API testing	Free	Free	Free tier
Device cloud (BrowserStack-type), optional	Optional	₹3,000	Planning
Security scanning: Dependabot + CodeQL (public/free tier)	Free	Free	Verify GitHub plan limits

One-time equipment	Class	₹	Basis
Mac for iOS builds/signing (planning allowance)	Paid	₹75,000	Quote needed
Test devices: 3 Android + 1 iPhone (planning allowance)	Paid	₹60,000	Quote needed

8.5 App-store and developer accounts

Item	USD	₹ at planning FX	Status
Google Play Console registration (one-time)	US$25	₹2,250	Last known US$25 - NOT verified this session
Apple Developer Program (annual)	US$99	₹8,910	Last known US$99/yr (INR price differs) - NOT verified
D-U-N-S number (organisation accounts)	-	-	Free from D&B via store flow - verify
Store assets: screenshots, icon, listing (in-house hours)	-	-	Covered in T20

Verify before quoting

Fees above are the last values the authors knew and were **not confirmed** on Apple/Google official pages in this edition. Open the Google Play Console help and Apple Developer Program pages, record the fee, currency, tax and access date in the evidence register. Also confirm current identity rules (organisation accounts need a D-U-N-S number; new personal Google accounts may face closed-testing prerequisites before production release). Account approval never guarantees app approval.

8.6 One-time infrastructure

Item	₹	Basis
Domain registration (1 yr)	₹1,000	Planning
DNS / SSL (managed, Let's Encrypt or provider)	₹0	Free tier
Email/SMS sender setup, DLT registration (India SMS)	₹6,000	Planning - verify telecom/provider fees
Initial load test & restore drill (tool/credits)	₹5,000	Planning
Legal: privacy policy, ToS, DPA templates review	₹25,000	Quote needed
Total one-time infrastructure	**₹37,000**	

8.7 Non-labour setup summary

Component	₹
Tooling for build period (paid, non-optional) ×5 months	₹25,000
Mac + test devices	₹1,35,000
One-time infrastructure	₹37,000
Store accounts, year 1	₹11,160
Total non-labour setup (cash)	**₹2,08,160**

Budget check

Cash need ₹2,08,160 vs intake budget ₹3,00,000 → within budget.

8.8 User-scale simulation

Driver	Low	Medium	High
MAU ÷ registered users	40%	60%	80%

Driver	Low	Medium	High
DAU ÷ MAU	25%	35%	45%
Sessions per DAU / day	2	3	4
API requests per session	15	25	35
Photos per DAU / day	3	6	10
OTP per MAU / month	2	2	2
Emails per MAU / month	6	10	14

```
Peak req/s = (DAU × sessions × req/session) ÷ (10 h × 3600) × peak factor 5
Backend instances = max(2, ceil(peak req/s ÷ 150))        (2 minimum for availability)
Database tier by DAU: ≤2k Small · ≤10k Medium · ≤40k Large · else XL + replica
Storage GB = photos/mo × 0.6 MB ÷ 1024 × 12 months retained
```

Registered	MAU	DAU	Peak concurrent	Peak req/s	API req/mo	Stored GB (12 mo)	Egress GB/mo	Model
100	60	21	1.6	0.2	0.05 M	26.6	3.7	1 Lean
500	300	105	7.9	1.1	0.24 M	132.9	18.4	1 Lean
1,000	600	210	15.8	2.2	0.47 M	265.8	36.8	2 Professional
5,000	3,000	1,050	78.8	10.9	2.36 M	1,328.9	184.1	2 Professional
10,000	6,000	2,100	157.5	21.9	4.72 M	2,657.8	368.3	2 Professional
25,000	15,000	5,250	393.8	54.7	11.81 M	6,644.5	920.7	2 Professional
50,000	30,000	10,500	787.5	109.4	23.62 M	13,289.1	1,841.4	3 Enterprise
1,00,000	60,000	21,000	1,575.0	218.8	47.25 M	26,578.1	3,682.8	3 Enterprise

8.9 Monthly production cost by scale (infrastructure only)

Registered users	Low usage ₹/mo	Medium ₹/mo	High ₹/mo	Medium ₹ per user
100	₹15,245	₹15,357	₹15,588	₹153.57
500	₹15,492	₹16,053	₹17,206	₹32.11
1,000	₹15,802	₹16,922	₹19,229	₹16.92
5,000	₹18,278	₹23,878	₹35,413	₹4.78
10,000	₹21,373	₹43,874	₹66,944	₹4.39
25,000	₹41,957	₹69,960	₹1,27,635	₹2.80
50,000	₹57,432	₹1,39,237	₹2,58,088	₹2.78
1,00,000	₹88,381	₹2,26,190	₹4,67,393	₹2.26

Why cost is not linear

Compute stays at two instances up to the largest simulated scale because the workload is light (peak req/s is small); cost steps at database, monitoring and storage tiers. These are model outputs from planning unit prices - run a load test before relying on instance counts, and refresh unit prices from official calculators.

Component (10,000 users, medium)	₹ / month
Backend compute	₹7,000
Database (Medium)	₹14,000
File storage	₹5,316
CDN / egress	₹2,946
Backups	₹2,729
SMS OTP	₹3,000
Email	₹4,800
Push (FCM)	₹0
Monitoring & logs	₹4,000
Domain & DNS	₹83

Component (10,000 users, medium)	₹ / month
Total infrastructure	**₹43,874**

8.10 Three infrastructure models

Model	One-time setup (infra + store yr-1 + extras)	Monthly	Annual	Capacity	Limitations
1 - Lean MVP	₹48,160	₹3,383	₹40,596	up to ~500 registered users, 1 region, no HA	Single point of failure; free-tier quotas can change; manual restore; no staging parity
2 - Professional	₹88,160	₹23,878	₹2,86,540	~1,000-25,000 registered users (see simulation)	Single region; RTO ~4 h, RPO ~15 min (PITR)
3 - Enterprise / Scalable	₹2,98,160	₹1,77,237	₹21,26,840	50,000+ registered users, multi-AZ, DR region	Needs dedicated DevOps; higher fixed cost

Model 2 monthly shown at 5,000 users (medium usage, ₹23,878); Model 3 at 50,000 users plus ₹38,000 for multi-AZ, DR copy, WAF, SSO and security tooling (planning). Setup extras: Model 2 +₹40,000 (staging, load test, hardening), Model 3 +₹2,50,000 (DR design, pen test).

Model 1 item	₹ / month
Compute: 1 small container/VM	₹1,500
Database: free/entry managed PG	₹800
Storage + CDN	₹300
Monitoring (free tiers)	₹0
Email/SMS (pay-per-use)	₹500
Backups: daily dump to object store	₹200
Domain/DNS	₹83
Total	**₹3,383**

Model	Scaling triggers	Risks	Upgrade requirements
1 - Lean MVP	DB CPU >60% for 1 week, p95 API >800 ms, >400 users, first paying B2B contract with SLA	Outage = full downtime; free-tier terms may change without notice	Move DB to managed HA, add second instance, add staging env
2 - Professional	Sustained peak RPS >70% of capacity; contractual 99.9% SLA; data-residency ask	Cost step-ups at tier boundaries; vendor lock-in on managed services	Multi-AZ DB, autoscaling, WAF, read replica
3 - Enterprise / Scalable	Enterprise contract needing 99.95% SLA, SSO, audit export, DR drills	Over-provisioning if demand does not arrive	Multi-region, SIEM, formal ISO 27001 / SOC 2 programme

> **Free tiers**
> Never assume a free tier, quota or term will remain unchanged. Record the date and page of every free-tier dependency and review quarterly.

11 Phase 9 - Pricing, profit & break-even

Gate 8

9.1 Custom development pricing

```
Project price = Labour + Setup + Third-party + Allocated overhead + Risk allowance + Target profit
= Total cost ÷ (1 − target margin)       [margin on price, not mark-up on cost]
```

Line	Client #1 (full build)	Client #2 (reusable tasks at 40% effort)
Development labour (expected)	₹15,00,549	₹10,56,604
Setup & third-party (non-labour)	₹2,08,160	₹2,08,160
Allocated company overhead (12% of labour)	₹1,80,066	₹1,26,792
Risk allowance (50% of Pess−Exp)	₹2,88,427	₹2,88,427
Total cost	₹21,77,202	₹16,79,984
Minimum price (margin 15%)	₹25,61,414	₹19,76,451
Target price (margin 30%)	**₹31,10,289**	**₹23,99,977**

Reusable share of expected labour: 49%. Customer-specific cost is recovered from the client; reusable IP stays company property and is licensed, not transferred. Quote client #1 at or above the minimum price only with founder approval; a quote below minimum margin is a Gate 8 rejection.

Check: at pessimistic labour ₹20,77,404 the client #1 target price ₹31,10,289 still earns 19% margin before the risk allowance is consumed.

9.2 White-label and enterprise

White-label	₹
Setup labour (branding, config, store submission)	₹1,00,012
Setup price = labour × (1+12%) ÷ (1−30%)	₹1,60,019
Annual licence + hosting (planning)	₹1,20,000

Enterprise = Business plan economics + custom integration labour (priced by the formula above) + 25% premium for SSO, audit export, SLA and security questionnaires; subscription from ₹24,999/month.

9.3 SaaS unit economics

```
Cost to serve (plan) = users × variable infra/user (₹1.88) + 2% payment fee + support minutes × ₹365/h ÷ 60
ARPA = Σ(mix × fee) × billing factor = ₹3,467 /customer/month   (billing factor 0.950: 30% pay annually for 10 months)
Gross margin/customer = ARPA − payment fee − variable infra − support = ₹3,241  (93.5%)
LTV = gross margin ÷ churn = ₹1,29,639   ;   LTV:CAC = 14.4   (CAC ₹9,000, churn 2.5%/mo)
```

9.4 Pricing tiers (ex-GST; 18% GST added on invoice)

Plan	Monthly	Annual (10 mo)	Users	Projects	Storage	API/day	Support	Cost to serve	Contribution	Margin
Starter	₹1,499	₹14,990	3	2	5 GB	5,000	10 min/mo	₹96	₹1,403	94%
Professional	₹3,999	₹39,990	10	10	25 GB	25,000	25 min/mo	₹251	₹3,748	94%
Business	₹9,999	₹99,990	30	Unlimited*	100 GB	100,000	60 min/mo	₹621	₹9,378	94%
Enterprise	from ₹24,999	custom	custom	custom	custom	custom	SLA	quote	quote	≥ 60% target

Overage	Rate (planning)
Extra user	₹150 / user / month
Extra storage	₹25 / GB / month
Extra API	₹40 per 10,000 requests/day band
Priority support	₹2,500 / incident-hour

*Unlimited projects under fair use. Prices are hypotheses until validated in Phase 1 interviews and the pilot.

9.5 Profitability by paying-customer count (monthly)

Paying cust.	Users	Revenue/mo	Hosting	Database	Storage, CDN, SMS, email, monitoring	Payment fees	Maintenance + roadmap eng.	Support	S&M + G&A
10	95	₹34,666	₹7,000	₹6,000	₹2,348	₹693	₹65,600	₹1,384	₹60,000
25	238	₹86,664	₹7,000	₹6,000	₹2,597	₹1,733	₹82,000	₹3,460	₹60,000
50	475	₹1,73,328	₹7,000	₹6,000	₹3,009	₹3,467	₹98,400	₹6,920	₹64,666
100	950	₹3,46,655	₹7,000	₹6,000	₹3,835	₹6,933	₹1,31,200	₹13,840	₹99,331
250	2,375	₹8,66,637	₹7,000	₹6,000	₹6,313	₹17,333	₹1,96,800	₹34,599	₹2,23,328
500	4,750	₹17,33,275	₹7,000	₹6,000	₹10,444	₹34,665	₹2,62,400	₹69,198	₹3,96,655
1,000	9,500	₹34,66,550	₹7,000	₹6,000	₹18,704	₹69,331	₹3,93,600	₹1,38,396	₹7,73,310

Paying cust.	Revenue/mo	Total cost	Operating profit	Op. margin	GST collected (not revenue)	Income tax	Profit after tax
10	₹34,666	₹1,43,025	-₹1,08,360	-313%	₹6,240	₹0	-₹1,08,360
25	₹86,664	₹1,62,790	-₹76,126	-88%	₹15,599	₹0	-₹76,126
50	₹1,73,328	₹1,89,461	-₹16,133	-9%	₹31,199	₹0	-₹16,133
100	₹3,46,655	₹2,68,139	₹78,516	23%	₹62,398	₹19,763	₹58,754
250	₹8,66,637	₹4,91,373	₹3,75,265	43%	₹1,55,995	₹94,454	₹2,80,811
500	₹17,33,275	₹7,86,362	₹9,46,913	55%	₹3,11,989	₹2,38,338	₹7,08,575
1,000	₹34,66,550	₹14,06,341	₹20,60,209	59%	₹6,23,979	₹5,18,555	₹15,41,654

Maintenance + roadmap engineering hours/mo by scale: 10: 80 h, 25: 100 h, 50: 120 h, 100: 160 h, 250: 240 h, 500: 320 h, 1000: 480 h at blended ₹820/h. S&M = max(₹30,000, 20% of revenue). Income tax 25.17% applies only to positive operating profit; verify with the company's CA. Revenue is never profit.

9.6 Break-even and sensitivity

```
Operating break-even at 70 paying customers (revenue ₹2,42,658/mo = costs ₹2,40,612/mo)
Payback of build labour ₹15,00,549 + setup ₹2,08,160 = ₹17,08,709 from 100-customer profit ₹78,516/mo ≈ 22 months; from 250-customer profit ₹3,75,265/mo ≈ 5 months
```

Sensitivity at 100 customers	Operating profit ₹/mo	Change vs base
Base case	₹78,516	-
Price -20%	₹24,438	-₹54,078
Price +20%	₹1,32,594	₹54,078
Usage: low	₹79,580	₹1,064
Usage: high	₹76,325	-₹2,192

Churn	LTV	LTV:CAC
1.5%/mo	₹2,16,065	24.0
2.5%/mo	₹1,29,639	14.4
4.0%/mo	₹81,024	9.0
6.0%/mo	₹54,016	6.0

9.7 Maintenance packages (Phase 13 pricing)

Package	Hours/mo	Labour	Third-party	Cost/mo	Margin	Price/mo	Response target	Channel	Includes
Basic	12 h	₹9,064	₹2,500	₹11,564	40%	**₹19,273**	Next business day	Email, 5×9	Critical bugs, OS compat patch 1x/qtr
Professional	36 h	₹27,192	₹6,000	₹33,192	40%	**₹55,320**	4 business hours (P1) / 1 day (P2)	Email + chat, 6×10	Monthly release, dependency & security updates
Enterprise	90 h	₹67,980	₹18,000	₹85,980	35%	**₹1,32,277**	1 hour (P1, 24×7 on-call) / 4 h (P2)	Phone + named engineer	Weekly releases, DR drill, SLA credits

```
Package cost = eng h × avg(Flutter, Backend rate) + QA h × QA rate + support h × support rate + third-party tools
Price = cost ÷ (1 − margin)
```

Commercial boundary

Hours above the package cap are billed at the blended rate plus 20%; unused hours do not roll over. Out of scope: new features, third-party outage, customer-caused data loss, OS versions older than the support matrix. P1 escalation: on-call engineer → tech lead → CTO within 30 / 60 / 120 minutes.

9.8 Commercial recommendation

Recommendation (sample app)	Value
Custom build - minimum / target price (client #1)	₹25,61,414 / ₹31,10,289
Custom build - target price (client #2, reuse)	₹23,99,977
White-label setup fee	₹1,60,019
White-label annual licence	₹1,20,000
SaaS list prices	Starter ₹1,499 / Professional ₹3,999 / Business ₹9,999 / Enterprise from ₹24,999
Annual subscription	10 × monthly fee (2 months free)
Monthly maintenance (post-handover)	Basic ₹19,273 / Professional ₹55,320 / Enterprise ₹1,32,277
Target gross margin	≥ 93% achieved (cost to serve); target operating margin ≥ 25% beyond break-even
Break-even customers	70
Expected operating profit at 100 / 250 / 500	₹78,516 / ₹3,75,265 / ₹9,46,913 per month

- Variable cost per customer is a small fraction of price, so margin grows with scale; the business is limited by engineering and acquisition cost, not hosting.
- Break-even at 70 customers is reachable with an annual acquisition plan (about 6 per month for 12 months); validate CAC in the pilot because LTV:CAC of 14 is only as good as the assumed ₹9,000 CAC.
- Custom-build price covers risk and company overhead, so a pessimistic outcome still leaves positive margin (check below).

- Maintenance packages are priced from real labour + third-party cost with a margin, and have hour caps - no unlimited support.

12 Phase 10 - Build & app-store publishing

Gate 10

OWNER
Release Manager + DevOps

INPUTS
Gate 7 and Gate 9 approvals

OUTPUT
Published, verified app

EXIT GATE
Gate 10

PROCEDURE

1. Run Android and iOS checklists in parallel.
2. Re-verify current store requirements on official pages on the submission day (policies, target API level, privacy rules change).
3. Submit only after Gate 10 sign-off. Do not promise an approval date to customers.
4. Run post-release verification within 2 hours of going live.

Android release

- ☐ Application ID / package name fixed (reverse-domain, never changes)
- ☐ versionName + monotonically increasing versionCode
- ☐ Icon, adaptive icon, splash
- ☐ Declare only needed permissions
- ☐ Generate upload key; enrol in Play App Signing; back up keystore offline (2 locations)
- ☐ Build signed AAB in CI
- ☐ Listing: title, short/full description, screenshots, feature graphic
- ☐ Data safety form matches SDK audit
- ☐ Privacy policy URL live
- ☐ Internal testing → closed/open testing as account type requires
- ☐ Submit for review; release staged (e.g. 10%→50%→100%)

iOS release

- ☐ Bundle identifier registered; certificates + provisioning managed (Xcode managed signing or fastlane match)
- ☐ Version + build number; entitlements (push, associated domains)
- ☐ Usage-description strings for every permission
- ☐ Archive in Xcode / CI macOS runner; upload via Xcode, Transporter or altool replacement
- ☐ App Store Connect record, screenshots per required device size, description, keywords
- ☐ App Privacy ('nutrition label') answers match SDK audit
- ☐ Demo account + review notes for App Review
- ☐ TestFlight internal then external beta
- ☐ Submit; respond to review within 24 h; phased release

Post-release verification

- ☐ Install from store on clean devices
- ☐ Register / login / password reset
- ☐ Main workflow end-to-end on production
- ☐ API connectivity to production only
- ☐ Payments (real small transaction, refunded)
- ☐ Push received on Android + iOS
- ☐ Crash reporting + analytics receiving events
- ☐ Support link and privacy policy open
- ☐ Store declarations reviewed against actual data flows

> **Common rejections**
> Missing demo credentials, privacy declarations that contradict SDK behaviour, broken links, crashes at launch, requesting permissions without clear purpose, in-app purchase rules for digital goods. Verify current store policy text before submission.

13 Phase 11 - Domain, DNS & deployment

Gate 9

OWNER
DevOps Engineer

INPUTS
Gate 8 and Gate 7

OUTPUT
Production environment

EXIT GATE
Gate 9

PROCEDURE

1. Follow the list in order; record evidence (screenshot or log) for each item.
2. Never put production secrets in the mobile app: anything shipped in an app can be extracted. Use per-user tokens; keep third-party keys server-side or restricted by package/bundle ID and quota.

- ☐ Register domain (ICANN-accredited registrar); enable registrar lock + 2FA
- ☐ Add DNS records **exactly as the hosting provider displays them** (A/AAAA/CNAME/TXT); never guess records
- ☐ Deploy backend container; set environment variables from secret manager
- ☐ Provision managed PostgreSQL; restrict to private network; enable automated backups + PITR
- ☐ Object storage with private buckets + signed URLs
- ☐ TLS certificate auto-renew; HSTS
- ☐ Run migrations through CI only (forward-only, reversible plan)
- ☐ Monitoring: uptime, error rate, p95 latency, queue depth, disk; alert routes to on-call
- ☐ Backup restore drill (documented RTO/RPO)
- ☐ Rollback: previous image tag + DB migration compatibility rule (expand/contract)
- ☐ Separate dev / staging / prod URLs, DBs and keys
- ☐ Production smoke test checklist signed by QA

14 Phase 12 - Showcase & sales

Gate 11

OWNER
Sales Executive + PM

INPUTS
Gate 8 pricing, working build

OUTPUT
Signed customer + onboarding

EXIT GATE
Gate 11

PROCEDURE

1. Prepare demo environment; rehearse; state limitations honestly.
2. Present pricing from the approved sheet only; deviations need founder approval.
3. Convert to pilot or subscription; execute SoW.
4. Handover and onboarding at Gate 11.

Showcase kit

- ☐ Working demo build + synthetic accounts (no real customer data)
- ☐ Seed data script
- ☐ Demo script (10 min): problem → 3 key workflows → admin view → limits → pricing
- ☐ Sales deck, one-page brochure, pricing sheet
- ☐ Proposal, Statement of Work, contract
- ☐ Onboarding guide, user manual, admin manual, training slides, handover document

Ownership and licensing

Topic	Default position (legal review required)
Source code	Client builds: ownership of customer-specific code transfers on full payment; reusable components remain company IP, licensed to the client
Third-party licences	Listed in THIRD_PARTY register with licence type; client receives the list
Customer data	Belongs to the customer; processor terms (DPA) define use, retention, deletion on exit
Limits & warranty	30-60 day defect warranty on frozen scope; no warranty on third-party outages or store decisions
Pilot	Paid pilot (e.g. 60 days) with success metrics and conversion price agreed upfront

15 Phase 13 - Maintenance & support

Prices and cost build-up are in §11 (9.7). Operational duties:

- ☐ Crash and ANR monitoring (daily review)
- ☐ Performance monitoring vs targets

☐ Security patches and dependency updates (monthly)
☐ New Android / iOS release compatibility (each beta season)
☐ Store compliance updates (target API level, privacy rules)
☐ API and DB maintenance (indexes, vacuum, migration hygiene)
☐ Backup verification monthly; restore drill quarterly
☐ Bug fixes by severity SLA
☐ Feature updates via CR
☐ Subscription renewals and version management (min supported version, forced-update flag)

Severity	Basic	Professional	Enterprise
P1 outage / data risk	Next business day	4 business hours	1 hour, 24×7 on-call
P2 major defect	3 days	1 day	4 hours
P3 minor	Backlog	Monthly release	Weekly release

16 Phase 14 - Employee management system

Role	Accountable for	Common failure to prevent
Business Researcher	Evidence register, market report	Unsourced numbers
Product Manager	PRD, CRs, budget, schedule, customer comms	Scope creep
UI/UX Designer	Prototype, tokens, usability results	Unapproved UI changes
Android / iOS / Flutter-RN Dev	Mobile code, unit/widget tests	Merging without review
Backend Developer	APIs, business logic, OpenAPI	Breaking API changes without version
Database Engineer	Schema, migrations, backups, performance	Manual prod changes
QA Engineer	Test plan, automation, release verdict	Passing with open Sev-1/2
DevOps Engineer	CI/CD, infra, monitoring, secrets	Secrets in repo
Sales Executive	Pipeline, demos, proposals	Off-price discounts
Customer Support Executive	Tickets, first-line triage, feedback	Promising unscoped work

In a small team one person may hold several roles; **hours are logged once** against the role actually performed, and the QA owner must not approve their own code.

Project record (required for every project)

Project record field	Example
Project ID	PRJ-2026-001
Project manager / tech lead / QA owner	named people
Developers assigned	named, with % allocation
Budget (approved)	₹17,08,709
Approved scope	PRD v1.0-frozen link
Estimated hours / actual hours	1,747 / (from time log)
Production cost	labour actual + non-labour actual
Selling price / expected margin	₹31,10,289 / 30%
Deadline / status	date / stage tracker value

Weekly report template

```
WEEKLY PERFORMANCE REPORT  (week ending ____)  Project: ______
Tasks planned / completed / carried over:   __ / __ / __
Hours estimated vs actual (variance %):     ____ / ____ ( __ %)
Defects opened / closed / reopened (by severity):
Delays (cause, days) and recovery plan:
Cost overrun to date (₹ and % of budget):
Customer feedback / NPS / tickets:
Risks and decisions needed from managers:
```

17 Phase 15 - Internal project portal

Build the portal as the company's own product using this OS: PRD first, then Gate-by-Gate. Roles: Employee (upload deliverables, log time), Manager (approve / reject / request revision), Finance, Admin.

1. Company dashboard
2. Project creation
3. Market research
4. Product requirements
5. Scope approval
6. UI/UX approval
7. Employee assignment
8. Development tracking
9. GitHub integration
10. Build management
11. QA tracking
12. Device testing
13. Production cost calculator
14. App pricing calculator
15. Cloud cost calculator
16. Android publishing tracker
17. iOS publishing tracker
18. Domain & DNS tracker
19. Customer management
20. Sales management
21. Subscription management
22. Maintenance tracker
23. Employee performance
24. Financial reports

Stage tracker rules

1. Each stage has required deliverables; the stage cannot close until all are uploaded.
2. Manager decision is one of Approve / Reject / Request revision, with comment; all decisions are audit-logged.
3. A rejected gate re-opens the stage and notifies the owner.

Core data model

Entity	Key fields
Project	id, name, type, stage, manager, tech_lead, qa_owner, budget, price, deadline
StageGate	project_id, gate (1-11), status (pending/approved/rejected/revision), approver, date, evidence[]
Deliverable	project_id, gate, file, uploaded_by, version
TimeEntry	project_id, task, role, user, hours, date
CostLine	project_id, category, estimated, actual, source_url, checked_on
Customer / Subscription	plan, MRR, start, renewal, churn_flag
Incident	project_id, severity, opened, resolved, root_cause
PriceSource	provider, page URL, value, currency, access date, reviewer

Dashboard KPIs

- Estimated vs actual project cost and variance
- Estimated vs actual monthly infrastructure
- Selling price, gross margin, MRR, active customers
- Pending tasks, delayed projects, production incidents

> **Suggested build**
> Phase-1 portal MVP in 6-8 weeks: projects, gates, deliverables, time entries, cost lines, price-source register. Calculators reuse the formulas in §10-§11.

18 Phase 16 - Sources & verification log

Official sources (primary references)

Source	URL
Android developer docs	https://developer.android.com/docs
Kotlin	https://kotlinlang.org/docs/home.html
Android Studio	https://developer.android.com/studio
Flutter	https://docs.flutter.dev/
React Native	https://reactnative.dev/docs/getting-started
Apple Developer documentation	https://developer.apple.com/documentation/
Apple Human Interface Guidelines	https://developer.apple.com/design/human-interface-guidelines/
App Store Connect help	https://developer.apple.com/help/app-store-connect/

Source	URL
Google Play Console help	https://support.google.com/googleplay/android-developer/
Firebase	https://firebase.google.com/docs
Supabase	https://supabase.com/docs
AWS	https://docs.aws.amazon.com/
Microsoft Azure	https://learn.microsoft.com/azure/
Google Cloud	https://cloud.google.com/docs
GitHub	https://docs.github.com/
OWASP MASVS	https://mas.owasp.org/MASVS/
OWASP MASTG	https://mas.owasp.org/MASTG/
NIST CSRC	https://csrc.nist.gov/
ICANN	https://www.icann.org/
MeitY	https://www.meity.gov.in/
CBIC GST	https://cbic-gst.gov.in/

Verification log - items that must be refreshed

Figure used in this document	Status	Action at Gate 8
Google Play registration fee	Last known - not verified	Read official Play Console help; record fee, currency, date
Apple Developer Program fee	Last known - not verified (page fetch did not show price)	Read developer.apple.com programs page
Cloud, SMS, email, storage unit prices	Planning assumptions	Use official calculators; paste output + date in PriceSource
SaaS tools (GitHub, Figma, Sentry...)	Planning	Check pricing pages
Income tax & GST rates	Planning - verify	Confirm with CA / CBIC notifications
Android vitals thresholds, store policies	Reference only	Re-read policy on submission day

> **Pricing rule**
> Every pricing calculation must cite the official provider page and the date checked. A price without source and date is not allowed into a quotation.

19 Appendix

Glossary

Term	Meaning
PERT	Three-point estimate (O + 4M + P) ÷ 6
ARPA	Average revenue per account per month
MAU/DAU	Monthly / daily active users
CAC / LTV	Customer acquisition cost / lifetime value (gross margin ÷ churn)
RTO / RPO	Recovery time / data-loss window objective
PWA	Progressive web app
MASVS / MASTG	OWASP Mobile verification standard / testing guide
CR	Change request

Day-1 action plan

1. Fill intake (§1) and confirm roles/rates (§10.1).
2. Create portal repo from template; apply Gate rules.
3. Start Phase 1 for SiteDiary (10 interviews in 2 weeks).
4. Open Google Play and Apple developer accounts early (identity checks take time).
5. Replace every Planning figure by verified source + date before Gate 8.

Structural / safety note

This OS contains no engineering-safety claims. For apps in regulated domains (health, finance, construction safety) obtain domain-expert and legal review; AI-generated output must be verified by a qualified person.

Planning figures are not quotations. Verify all provider prices and legal/tax positions before use. · Generated by build/build.py from model.py.